北京交通大學

信号与系统

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and Information Engineering

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2025.3

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Complex frequency domain analysis of continuous system responses

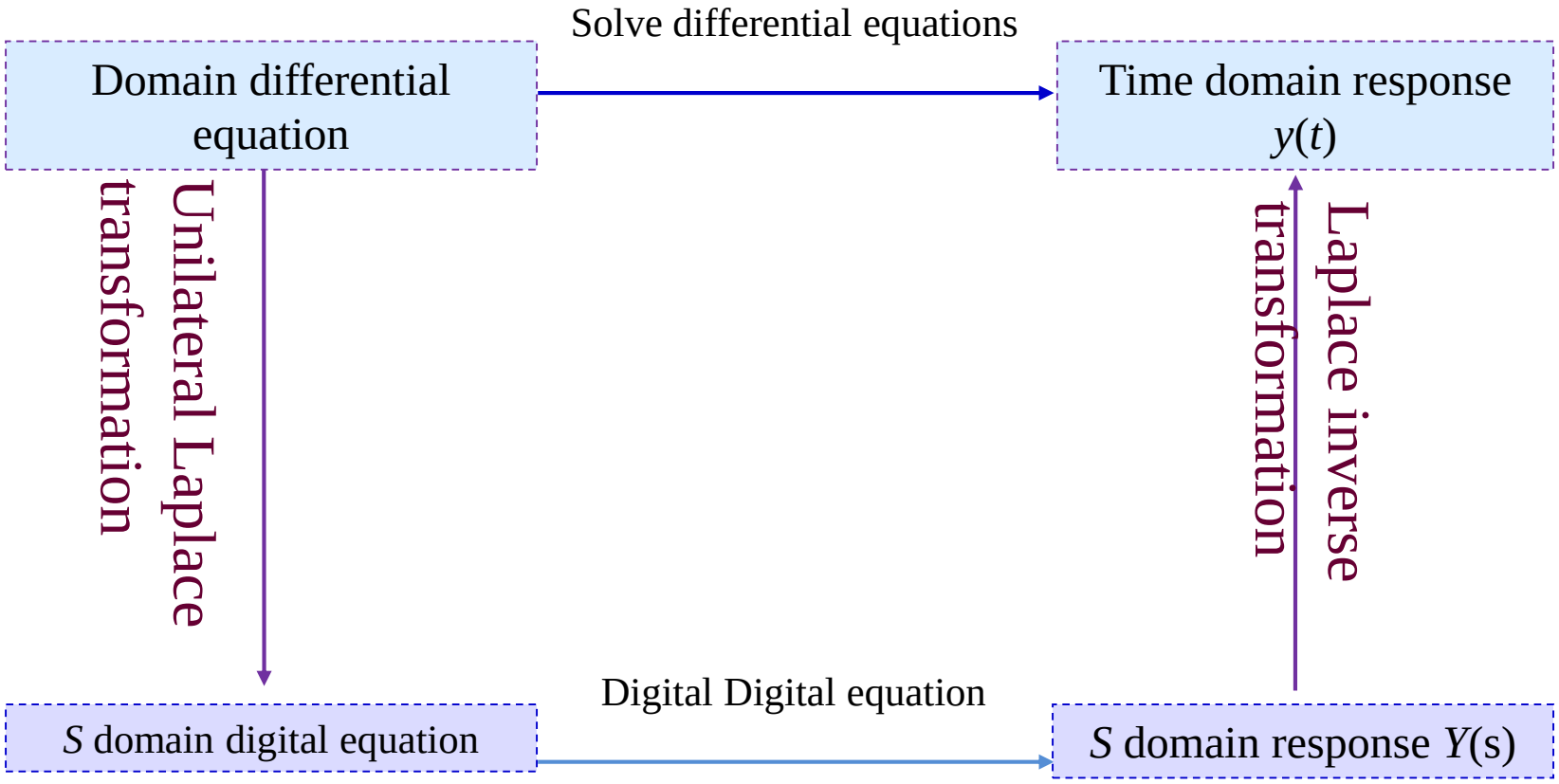
Consecutive signal analysis of frequency domain

- Reproduction domain representation of continuous signals
- Commonly used unilateral Laplas transformation
- Unilateral Laplace transformation nature
- Unilateral Laplace Reverse Conversion
- Bilateral Laplace transformation and inverse transformation

Consecutive LTI system complex frequency domain analysis

- Reproduction domain description of the continuous system
- System function and system characteristics
- Simulation of continuous time system
- Complex frequency domain analysis of system response
- Matlab implementation

Complex frequency domain analysis of system response



Complex frequency field analysis of system response

Example: S domain solution of the second -order system response

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+5\frac{\mathrm{d}y(t)}{\mathrm{d}t}+6y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+8x(t)$$

known $x(t) = \mathrm{e}^{-t}u(t)$, $y(0^-) = 3$, $y'(0^-) = 2$, solve $y(t)$。

Solve the steps:

☼Transform from Rahi transform the time domain differential equation into the S domain digital equation

☼ Solve the digital equation of the S domain, find $Y_{zi}(s)$, $Y_{zs}(s)$

☼Laplace inverse transformation, seeking the response time domain indication formula

Complex frequency field analysis of system response

Example:

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+5\frac{\mathrm{d}y(t)}{\mathrm{d}t}+6y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+8x(t)$$

Known $x(t)=\mathrm{e}^{-t}u(t)$, $y(0^-)=3$, $y'(0^-)=2$, solve $y(t)$。

Solution:

$$y''(t) \qquad 5y'(t) \qquad 6y(t)$$

$$\updownarrow \qquad \updownarrow \qquad \updownarrow$$

$$[s^2Y(s)-sy(0^-)-y'(0^-)] \qquad +5[sY(s)-y(0^-)] \qquad +6Y(s)$$

$$=2sX(s)+8X(s) \longleftarrow 2x'(t)+8x(t)$$

$$Y(s)=\underbrace{\frac{sy(0^-)+y'(0^-)+5y(0^-)}{s^2+5s+6}}_{Y_{zi}(s)}+\underbrace{\frac{2s+8}{s^2+5s+6}X(s)}_{Y_{zs}(s)}$$

Complex frequency field analysis of system response

Example:

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+5\frac{\mathrm{d}y(t)}{\mathrm{d}t}+6y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+8x(t)$$

Known $x(t)=\mathrm{e}^{-t}u(t)$, $y(0^-)=3$, $y'(0^-)=2$, solve $y(t)$。

Solution:

$$Y_{\mathrm{zi}}(s)=\frac{3s+17}{s^2+5s+6}=\frac{11}{s+2}-\frac{8}{s+3}$$

$$y_{\mathrm{zi}}(t)=\mathscr{L}^{-1}\{Y_{\mathrm{zi}}(s)\}=11\mathrm{e}^{-2t}-8\mathrm{e}^{-3t},\quad t\geq 0$$

Complex frequency field analysis of system response

Example:

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+5\frac{\mathrm{d}y(t)}{\mathrm{d}t}+6y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+8x(t)$$

Known $x(t)=\mathrm{e}^{-t}u(t)$, $y(0^-)=3$, $y'(0^-)=2$, solve $y(t)$。

Solution:

$$Y_{\mathrm{zs}}(s)=\frac{2s+8}{s^2+5s+6}\cdot\frac{1}{s+1}=\frac{2s+8}{(s+2)(s+3)}\cdot\frac{1}{s+1}$$

$$=\frac{3}{s+1}-\frac{4}{s+2}+\frac{1}{s+3}$$

$$y_{\mathrm{zs}}(t)=\mathscr{L}^{-1}\{Y_{\mathrm{zs}}(s)\}=(3\mathrm{e}^{-t}-4\mathrm{e}^{-2t}+\mathrm{e}^{-3t})\cdot u(t)$$

$$y(t)=y_{\mathrm{zi}}(t)+y_{\mathrm{zs}}(t)=3\mathrm{e}^{-t}+7\mathrm{e}^{-2t}-7\mathrm{e}^{-3t},t\geq 0$$

Complex frequency field analysis of system response

[Example1] $$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+7\frac{\mathrm{d}y(t)}{\mathrm{d}t}+12y(t)=x(t)$$

Known $x(t)=u(t)$, $y(0^-)=1$, $y'(0^-)=2$, solve $y(t)$。

[Answer] $$y_{\mathrm{zi}}(t)=6\mathrm{e}^{-3t}-5\mathrm{e}^{-4t}, t\geq 0$$

$$y_{\mathrm{zs}}(t)=(\frac{1}{12}-\frac{1}{3}\mathrm{e}^{-3t}+\frac{1}{4}\mathrm{e}^{-4t})u(t)$$

$$y(t)=\frac{1}{12}+\frac{17}{3}\mathrm{e}^{-3t}-\frac{19}{4}\mathrm{e}^{-4t}, t\geq 0$$

Complex frequency field analysis of system response

[Example2] $$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+4\frac{\mathrm{d}y(t)}{\mathrm{d}t}+4y(t)=3\frac{\mathrm{d}y(t)}{\mathrm{d}t}+2x(t)$$

Known $x(t) = 4u(t)$, $y(0^-) = -2$, $y'(0^-) = 3$, solve $y(t)$。

[Answer] $$y_{\mathrm{zi}}(t)=-2\mathrm{e}^{-2t}-t\mathrm{e}^{-2t},\quad t\geq 0$$

$$y_{\mathrm{zs}}(t)=(0.5+2t\mathrm{e}^{-2t}-0.5\mathrm{e}^{-2t})u(t)$$

$$y(t)=0.5-2.5\mathrm{e}^{-2t}+t\mathrm{e}^{-2t},\quad t\geq 0$$

Complex frequency field analysis of system response

[Example3] $$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+4\frac{\mathrm{d}y(t)}{\mathrm{d}t}+8y(t)=3\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

Known $x(t)=\mathrm{e}^{-t}u(t)$, $y(0^-)=5$, $y'(0^-)=3$, solve $y(t)$。

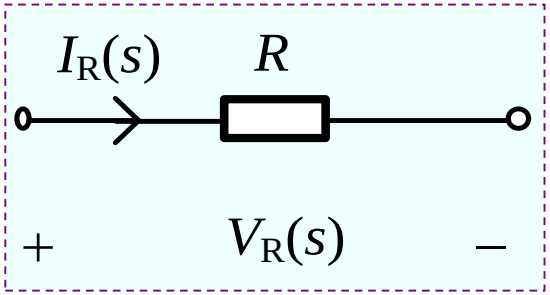
[Answer] $$y_{\mathrm{zi}}(t)=5\mathrm{e}^{-2t}\cos(2t)+\frac{13}{2}\mathrm{e}^{-2t}\sin(2t),\quad t\geq 0$$

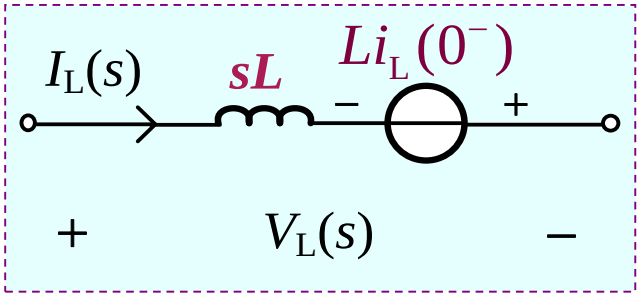
$$y_{\mathrm{zs}}(t)=\left[-0.4\mathrm{e}^{-t}+0.4\mathrm{e}^{-2t}\cos(2t)+1.7\mathrm{e}^{-2t}\sin(2t)\right]u(t)$$

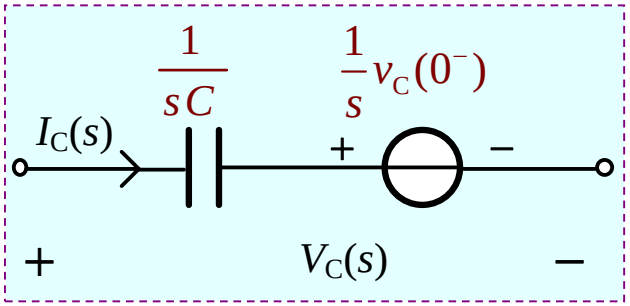
$$y(t)=-0.4\mathrm{e}^{-t}+5.4\mathrm{e}^{-2t}\cos(2t)+8.2\mathrm{e}^{-2t}\sin(2t),\quad t\geq 0$$

Complex frequency field analysis of system response

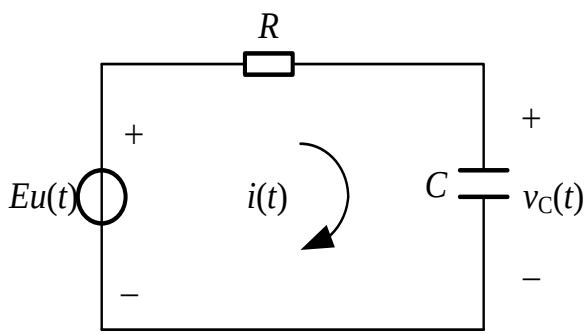
R、L、C Complex frequency domain model

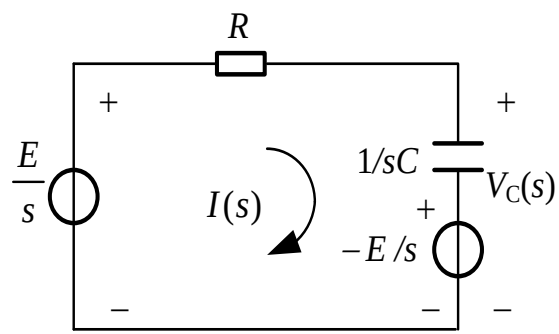






Example: The initial state of the icon is $v_C(0^-) = -E$, and the voltage $v_C(t)$ at both ends of the capacitance is found.





Solution: Establish a Complex frequency domain model for circuit

Write the loop equation from the frequency frequency domain model

$$(R+\frac{1}{sC})I(s)=\frac{E}{s}+\frac{E}{s}$$

Seeking the circuit current

$$I(s)=\frac{2E}{s(R+\frac{1}{sC})}$$

Capacitor voltage is

$$V_C(s)=\frac{I(s)}{sC}-\frac{E}{s}=E(\frac{1}{s}-\frac{2}{s+\frac{1}{RC}})$$

$$v_C(t)=E(1-2e^{-\frac{1}{RC}t}),t\geq 0$$

Comprehensive question: Describe the differential equation of the LTI causality system for a certain period of time is

$$y''(t)+7y'(t)+10y(t)=2x'(t)+3x(t)$$

It is known $x(t)=\mathrm{e}^{-t}u(t)$,$y(0^-)=1$, $y'(0^-)=1$

solved from the frequency frequency domain:

(1) zero input response $y_{zi}(t)$, zero state response $y_{zs}(t)$, and fully respond to $y(t)$.

(2) The system function $H(s)$, the unit's impulse response $h(t)$, and determine whether the system is stable.

(3) Draw the direct simulation frame diagram of the system.

Comprehensive question: The differential equation describing a continuous-time LTI causal system is $y''(t)+7y'(t)+10y(t)=2x'(t)+3x(t)$

Known $x(t)=\mathrm{e}^{-t}u(t)$ $y(0^-)=1, y'(0^-)=1$,solved from the complex frequency domain:

(1) Zero input response $y_{zi}(t)$, zero state response$y_{zs}(t)$, complete response $y(t)$.

Solution: (1) Perform unilateral Laplace transform on both sides of the differential equation to get

$$s^2Y(s)-sy(0^-)-y'(0^-)+7sY(s)-7y(0^-)+10Y(s)=(2s+3)X(s)$$

$$Y(s)=\frac{sy(0^-)+y'(0^-)+7y(0^-)}{s^2+7s+10}+\frac{2s+3}{s^2+7s+10}X(s)$$

The complex frequency domain expression of the zero-input response is

$$Y_{zi}(s)=\frac{s+8}{s^2+7s+10}=\frac{2}{s+2}+\frac{-1}{s+5}$$

One-sided inverse Laplace transform can be obtained

$$y_{zi}(t)=\mathscr{L}^{-1}\{Y_{zs}(s)\}=2\mathrm{e}^{-2t}-\mathrm{e}^{-5t}, t\geq 0$$

Comprehensive question: The differential equation describing a continuous-time LTI causal system is $y''(t)+7y'(t)+10y(t)=2x'(t)+3x(t)$

Known $x(t)=\mathrm{e}^{-t}u(t)$ $y(0^-)=1, y'(0^-)=1$,solved from the complex frequency domain:

(1) Zero input response $y_{zi}(t)$, zero state response$y_{zs}(t)$, complete response $y(t)$.

Solution: $$Y(s)=\frac{sy(0^-)+y'(0^-)+7y(0^-)}{s^2+7s+10}+\frac{2s+3}{s^2+7s+10}X(s)$$

The complex frequency domain expression of the zero-state response is

$$Y_{zs}(s)=\frac{2s+3}{(s^2+7s+10)(s+1)}=\frac{1/4}{s+1}+\frac{1/3}{s+2}+\frac{-12/7}{s+5}$$

Perform the inverse Laplace transform to get

$$y_{zs}(t)=\mathscr{L}^{-1}\{Y_{zs}(s)\}=(\frac{1}{4}\mathrm{e}^{-t}+\frac{1}{3}\mathrm{e}^{-2t}-\frac{7}{12}\mathrm{e}^{-5t})u(t)$$

The full response is

$$y(t)=y_{zi}(t)+y_{zs}(t)=\frac{1}{4}\mathrm{e}^{-t}+\frac{7}{3}\mathrm{e}^{-2t}-\frac{19}{12}\mathrm{e}^{-5t}, t>0$$

Comprehensive question: The differential equation describing a continuous-time LTI causal system is $y''(t)+7y'(t)+10y(t)=2x'(t)+3x(t)$

Known $x(t)=\mathrm{e}^{-t}u(t)$ $y(0^-)=1, y'(0^-)=1$,solved from the complex frequency domain:

(2) System function $H(s)$, unit impulse response $h(t)$, and judge whether the system is stable.

Solution: $$Y_{zs}(s)=\frac{2s+3}{(s^2+7s+10)}X(s)$$

(2) According to the definition of the system function, we can get

$$H(s)=\frac{Y_{zs}(s)}{X(s)}=\frac{2s+3}{s^2+7s+10}=\frac{-1/3}{s+2}+\frac{7/3}{s+5}$$

Perform inverse Laplace transform to get

$$h(t)=\mathscr{L}^{-1}[H(s)]=(-\frac{1}{3}\mathrm{e}^{-2t}+\frac{7}{3}\mathrm{e}^{-5t})u(t)$$

For the causal system, the pole of the system function is -2, and -5 is located in the left half s-plane, so the system is stable.

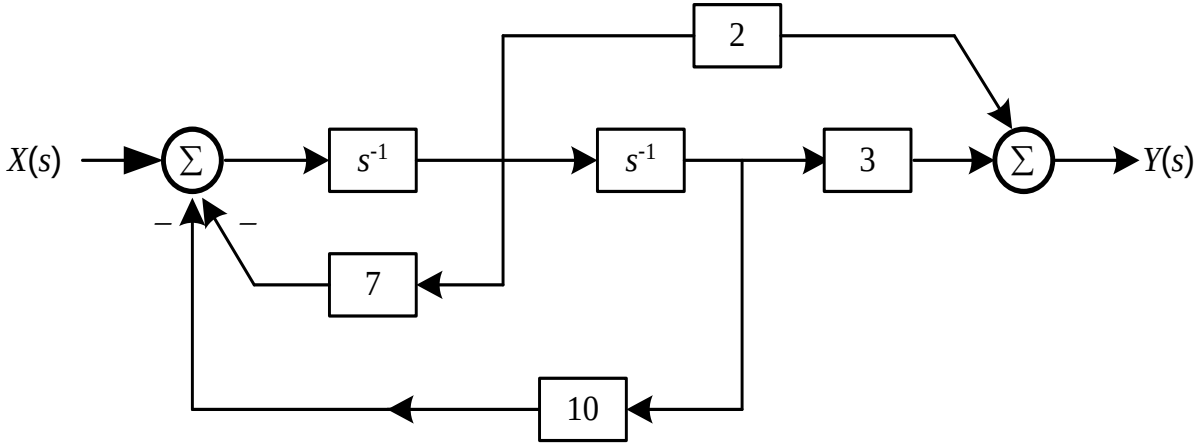
Comprehensive question: The differential equation describing a continuous-time LTI causal system is $y''(t)+7y'(t)+10y(t)=2x'(t)+3x(t)$

Known $x(t)=\mathrm{e}^{-t}u(t)$ $y(0^-)=1, y'(0^-)=1$,solved from the complex frequency domain：

(3) Draw the direct simulation frame diagram of the system.

Solution： (3) Repair the system function into

$$H(s)=\frac{2s^{-1}+3s^{-2}}{1+7s^{-1}+10s^{-2}}$$



Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!